

Reformulation and Solution of the Master Equation for Multiple-Well Chemical Reactions

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ABSTRACT: We consider an alternative formulation of the master equation for complex-forming chemical reactions with multiple wells and bimolecular products. Within this formulation the dynamical phase space consists of only the microscopic populations of the various isomers making up the reactive complex, while the bimolecular reactants and products are treated equally as sources and sinks. This reformulation yields compact expressions for the phenomenological rate coefficients describing all chemical processes, i.e., internal isomerization reactions, bimolecular-to-bimolecular reactions, isomer-to-bimolecular reactions, and bimolecular-to-isomer reactions. The applicability of the detailed balance condition is discussed and confirmed. We also consider the situation where some of the chemical eigenvalues approach the energy relaxation time scale and show how to modify the phenomenological rate coefficients so that they retain their validity.

$$\frac{d}{dt}|f\rangle = -\hat{G}|f\rangle + \sum_v s_v(t)|p^{(v)}\rangle$$
$$k_{v \rightarrow \mu} = \frac{1}{Q_v} \sum_{\lambda} \frac{1}{\Lambda_{\lambda}} p_{\lambda}^{(\mu)} p_{\lambda}^{(v)}$$

I. INTRODUCTION

The role of theoretical kinetics in global chemical modeling is gradually undergoing a paradigm shift. Historically, theory has served primarily as a means for interpolating and extrapolating experimental data. Recent advances in theoretical methodologies have led to predictions whose accuracy can rival the accuracy of many experimental studies. Correspondingly, theoretical elementary reaction kinetics has become an important tool for phenomenological modeling in chemical kinetics.¹ When experimental data is available, the best kinetic estimates are generally obtained from careful comparisons of the experimental data with theoretical predictions, while paying close attention to the uncertainty limits for both.²

Global models for chemical processes require phenomenological rate coefficients describing the rates of individual elementary reactions over broad ranges of temperature and pressure. The derivation of such rate coefficients is complicated by the need to consider the coupling of collision induced energy transfer with the microscopic chemical transformations that are continually occurring for all sufficiently excited species. Furthermore, these chemical and energetic transformations generally occur over complex potential energy surfaces representing multiple chemical isomers and multiple sets of bimolecular reactants and products. In the time-dependent master equation (ME) approach, one adopts a stochastic description and develops explicit expressions for the transition probabilities of the collisional and chemical processes. By directly solving the ME, energy-dependent microscopic populations for all distinct chemical species can be obtained as functions of time.

Alternatively, one can try to relate the ME to the phenomenological rate coefficients that describe the changes of the macroscopic populations (concentrations) of the different chemical species. For reactions involving a reactive complex with a single well and a single set of bimolecular species, various procedures for doing so are outlined in detail in some leading monographs on theoretical kinetics.^{3,4} These

descriptions include generalization to the case of multiple product channels.

Two distinct perspectives have been employed in ME studies of reactions involving a single reactive complex. One perspective focused on the kinetics of just the reactive complex, with the bimolecular reactants treated as a decoupled source term, which was left undefined.^{3–7} An alternative perspective added the bimolecular reactants to the population vector and explicitly provided the rate equation for the reactant concentrations including reversible dissociation and association processes.⁸ While the first perspective was more convenient for relating the ME to the phenomenological rate coefficients, the second one allowed for the direct microscopic modeling of specific systems. The explicit treatment of the concentrations of the bimolecular reactants allowed for the explicit generation of the full time-dependent macroscopic populations for all chemical species. From these populations, phenomenological rate coefficients could then be derived via an experimental approach involving the search for conditions with exponential population decays. Both perspectives have proven valuable in understanding the kinetics of single well systems. Useful descriptions of the relationships between these two approaches were provided in the work of Hanning-Lee et al.⁸ as well as in our discussion of detailed balance in association/dissociation reactions.⁹

At least at combustion temperatures, many, if not most, reactions occur over more complex potential energy surfaces that involve the isomerization between multiple wells representing distinct chemical isomers. For such multiple-well reactions, the derivation of the phenomenological rate coefficients from the time-dependent populations is consid-

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erably more complex. Notably, the second perspective was readily extended to the treatment of such multiple-well systems.^{10–12} In contrast, the first perspective has not been generalized to the treatment of these systems.

The time evolution of the reactive complex is commonly described in terms of the eigenstates of the properly discretized kinetic relaxation operator. A major further advance came with the realization that there is a one-to-one correspondence between the relaxation of the small subset of the eigenstates with low eigenvalues, which we call chemically significant eigenstates (CSE),¹³ and the chemical kinetics described in terms of the phenomenological rate coefficients.^{13–17} The CSEs have eigenvalues that are well separated from the rest of the eigenvalues. Bartis and Widom¹⁸ were the first to recognize the relationship between CSEs and phenomenological rate coefficients. Their work, however, has been long underappreciated. This one-to-one correspondence allows for the derivation of simple explicit relationships between the eigenvectors and eigenvalues and the rate coefficients. Other approaches for deriving the phenomenological rate coefficients either implement simplifying assumptions^{19,20} or require the experimental perspective that relates exponential decays of species populations to rate coefficients.^{21,22} The latter approach encounters great difficulties at high temperatures where the decreasing separation between different chemically significant eigenvalues leads to overlapping contributions to the time decays. The CSE based expressions are particularly valuable in removing this high temperature ambiguity. An alternative CSE based derivation of Robertson et al.²³ demonstrated analytically that the rate constants are not dependent on initial conditions. Miller et al.²⁴ clarified the role of detailed balance in multiple-well reactions, while correcting some shortcomings in the earlier comments of Robertson et al.²³ on this topic.

Notably, this realization of the one-to-one correspondence between CSE relaxation and phenomenological rate equations also provides an opportunity for generalizing the first perspective from single-well to multiple-well systems. This study explores this generalization, first deriving explicit relationships between the eigenstates and the phenomenological rate coefficients and then discussing some of the ramifications of the results obtained. This new perspective provides a useful complement to that of our original derivation, just as the two perspectives proved useful in the early explorations for single-well reaction systems. In particular it allows one to obtain at once the full set of the phenomenological rate coefficients, including bimolecular-to-bimolecular, isomer-to-bimolecular, bimolecular-to-isomer, and isomer-to-isomer rate coefficients. Interestingly, this derivation of the phenomenological rate coefficients does not require the pseudo-first order assumption for the bimolecular-to-isomer microscopic rate coefficients, where one of the bimolecular reactants is assumed to be in a great excess over the other. This is especially important for self-reactions, where both bimolecular reactants are actually the same species. The formulation has some elements in common with the virtual components methodology of Knyazev and Tsang,²⁵ who have applied a single-well master equation model to describe the nonsteady state kinetics of the reactive complex on the time scale comparable to or faster than the energy relaxation one. The present approach is in some ways a generalization to multiple wells of the single-well methodology of Smith et al.⁶

Within the present alternative formulation the master equation is written in terms of the energy-dependent

microscopic populations for each isomer of the reactive complex, while each of the sets of bimolecular reactants is considered as a decoupled external source. The concentrations of bimolecular reactants are left undefined in the sense that the equations describing their time evolution are not considered. Nevertheless, we are able to derive a full set of phenomenological rate coefficients via our understanding of the one-to-one correspondence between the CSEs relaxation and the phenomenological rate equations. The coupled treatment for the bimolecular reactant concentrations and the reactive-complex isomer concentrations should be considered in the next stage of chemical modeling, where the phenomenological rate coefficients obtained from the current treatment would be used. This approach is valid as long as the time scale for variation of the bimolecular reactants concentrations is much longer than the collisional-energy relaxation time scale, which is generally true. The present formulation will be seen to yield simple expressions for all of the phenomenological rate coefficients of relevance to a given potential energy surface.

We present this alternative formulation of the multiple-well master equation in section II. Then, in section III, we provide the derivation of the phenomenological rate coefficients within this formalism. The extent to which the detailed balance condition is satisfied is an important aspect of any kinetic phenomenology.^{6,9,15,24} In section IV we provide a discussion of this condition within the present formalism and particularly how it relates to the approach of some of the CSE eigenvalues to the near continuum of energy relaxation eigenvalues. Some concluding remarks are then provided in section V.

II. MASTER EQUATION FORMULATION

We assume that there are N configurations (isomers) of the reactive complex, which are separated from each other and from the bimolecular species by sufficiently high barriers that they are chemically distinct. We further assume that for each isomer the internal relaxation is so fast that its internal state can be described in terms of internal integrals of motion only, which are the internal energy E and the angular momentum J , while all other coordinates and momenta are in statistical equilibrium. The linear momentum of the reactive complex as a whole is also an integral of motion. However, its relaxation is decoupled from internal energy and angular momentum relaxation. In the following discussion, for the sake of notational simplicity and numerical efficiency, we consider a simplified microcanonical approach and assume that the angular momentum distribution is the equilibrium one. The E, J -resolved generalization, when it is warranted, is straightforward.

The nonequilibrium, instantaneous statistical state of the reactive complex is then described in terms of the set of N distribution functions $f_i(E)$, $i = 1, \dots, N$, which are the microscopic populations for the i th isomer with energy E . They are considered as components of the vector $|f\rangle$ that represents the statistical state of the reactive complex. The master equation, which describes the time evolution of the nonequilibrium vector $|f\rangle$, can be symbolically written as,

$$\frac{d}{dt}|f\rangle = -\hat{G}|f\rangle + \sum_{\nu} s_{\nu}(t)|p^{(\nu)}\rangle \quad (1)$$

where $\hat{G}$ is the kinetic relaxation operator within the reactive complex and $|p^{(\nu)}\rangle$ is the time-independent vector describing microscopic population gain from the ν th pair of bimolecular reactants. Here we assume that the energy distribution of

bimolecular reactants is the thermal one and, therefore, their microscopic population time-dependence comes through the reactants macroscopic concentrations only. Explicit expressions for $p^{(\nu)}$ are given below. The quantity $s_\nu(t)$ is given by

$$s_\nu(t) = n_A^{(\nu)}(t)n_B^{(\nu)}(t)/Q_\nu \quad (2)$$

where $n_{A,B}^{(\nu)}(t)$ are the concentrations of the ν th bimolecular reactants (A and B), which explicitly depend on time. The quantity Q_ν is the standard partition function for the relative and internal motions of the reactants, which may generally be written as

$$Q_\nu = Q_A^{(\nu)}Q_B^{(\nu)}(2\pi)^{-3/2} \left(\frac{m_A^{(\nu)}m_B^{(\nu)}T}{m_A^{(\nu)} + m_B^{(\nu)}} \right)^{3/2} \quad (3)$$

Here $m_{A,B}^{(\nu)}$ and $Q_{A,B}^{(\nu)}$ are the masses and partition functions of the bimolecular fragments, respectively. Bimolecular reactants serve as a source for formation of the reactive complex. The reason for presenting this source in the form of eq 1 will be made clear later. Throughout this article atomic units are used, and the temperature T is measured in energy units.

The kinetic relaxation operator $\hat{G}$ consists of two parts: the collisional energy relaxation operator $\hat{Z}$ and the chemical isomerization and dissociation operator $\hat{K}$:

$$\hat{G} = \hat{Z} + \hat{K} \quad (4)$$

The collisional energy relaxation operator describes the transitions between different energy levels of the same isomer (vertical transitions), which are caused by nonreactive collisions with buffer gas molecules. It is usually described in terms of a model integral operator kernel $Z_i(E \rightarrow E')$ that has a relatively simple analytical form. It is also usually assumed that the collision frequency $\omega_i(E)$,

$$\omega_i(E) = \int dE' Z_i(E \rightarrow E') \quad (5)$$

is energy E and isomer i independent, $\omega_i(E) \equiv \omega$. These assumptions, which we implement, are not critical to the subsequent analysis. Their usage presumes that the result of the calculation is not very sensitive to the details of the collisional energy relaxation model. It is, probably, even more correct to say that information concerning $Z(E \rightarrow E')$ is so severely limited that a more detailed treatment is not warranted.

The principal feature of the collisional energy relaxation kernel $Z_i(E \rightarrow E')$ is that it should satisfy the microscopic detailed balance condition:

$$Z_i(E \rightarrow E')f_i^{(0)}(E) = Z_i(E' \rightarrow E)f_i^{(0)}(E') \quad (6)$$

where

$$f_i^{(0)}(E) = \rho_i(E) e^{-E/T} \quad (7)$$

is the Boltzmann distribution and $\rho_i(E)$ is the density of states for the i th isomer with energy E .

The chemical isomerization operator $\hat{K}$ describes the microscopic chemical transformations between different isomers and the dissociation of the reactive complex into different bimolecular products. It describes processes occurring in the absence of collisions, where the total internal energy E is conserved (horizontal transitions). At the E -resolved level it is expressed in terms of energy-dependent microcanonical rate coefficients $k_{i \rightarrow j}(E)$. RRKM theory provides an efficient tool for their calculation:

$$k_{i \rightarrow j}(E) = \frac{1}{2\pi} \frac{N_{ij}^\#(E)}{\rho_i(E)} \quad (8)$$

where $N_{ij}^\#(E)$ is the variationally optimized number of states for the transition state separating the initial i and final j chemical species. While the use of RRKM theory for the calculation of $N_{ij}^\#(E)$ is very convenient, and in most situations sufficient, eq 8 is more general. Reinterpreting $N_{ij}^\#(E)$ as the cumulative reaction probability²⁶ allows one to take into account transition state recrossings and barrier tunneling as well.

Using RRKM theory one readily obtains the following expression for the i th component of the bimolecular source vector $p^{(\nu)}$ in eq 1:

$$p_i^{(\nu)}(E) = \frac{1}{2\pi} N_{i,\nu}^\#(E) e^{-E/T} \quad (9)$$

where $N_{i,\nu}^\#(E)$ is the variationally optimized number of states for the transition state separating the i th reactive complex isomer and ν th bimolecular reactants.

Notably, it follows from eq 8 that the chemical isomerization operator $\hat{K}$ satisfies the microscopic detailed balance condition as does the collisional energy relaxation operator $\hat{Z}$

$$k_{i \rightarrow j}(E)\rho_i(E) = k_{j \rightarrow i}(E)\rho_j(E) \quad (10)$$

As a result, the kinetic relaxation operator $\hat{G}$ is self-adjoint

$$\langle g | \hat{G} | f \rangle = \langle f | \hat{G} | g \rangle \quad (11)$$

provided that the scalar product of two state vectors $|g\rangle$ and $|f\rangle$ is defined as follows:

$$\langle g | f \rangle = \sum_i \int dE f_i(E) g_i(E) / f_i^{(0)}(E) \quad (12)$$

where $f_i^{(0)}(E)$ is the Boltzmann distribution for the i th isomer of the reactive complex, eq 7.

An important feature of $\hat{G}$ is that it is a positive definite. This feature follows from the fact that it can be represented as an (infinite) sum of non-negatively defined operators representing transitions between arbitrary pairs of states both for chemical isomerization and collisional energy relaxation. As a result, all eigenvalues of $\hat{G}$ are non-negative. In the degenerate case, where one group of isomers is not coupled chemically with other isomers or with bimolecular species, the Boltzmann distribution for this group of isomers becomes an eigenvector of $\hat{G}$ with zero eigenvalue. We, however, will assume that all isomers are chemically coupled. Thus, without restriction of generality one can assume that all eigenvalues of $\hat{G}$ are positive.

To proceed we will assume that all eigenstates of $\hat{G}$ are known and expand an arbitrary state vector $|f\rangle$ in terms of the $|\lambda\rangle$ eigenvectors

$$|f\rangle = \sum_\lambda f_\lambda |\lambda\rangle \quad (13)$$

Then, multiplying eq 1 to the left with an arbitrary eigenvector $|\lambda\rangle$ and making use of eqs 7, 9, and 12, one obtains the following equation for the expansion coefficients f_λ

$$\frac{df_\lambda}{dt} = -\Lambda_\lambda f_\lambda + \sum_\nu s_\nu(t) p_\lambda^{(\nu)} \quad (14)$$

In this expression, Λ_λ is the eigenvalue associated with the $|\lambda\rangle$ eigenvector

$$p_{\lambda}^{(\nu)} \equiv \langle \lambda | p^{(\nu)} \rangle = \frac{1}{2\pi} \sum_i \int dE f_i^{(\lambda)}(E) N_{i,\nu}^{\#}(E) / \rho_i(E) \quad (15)$$

and $f_i^{(\lambda)}(E)$ is the i th component of the $|\lambda\rangle$ eigenvector.

III. PHENOMENOLOGICAL RATE COEFFICIENTS

Equation 14 allows one to obtain a formal solution for the master equation provided that the time dependence of the bimolecular reactant concentrations, which come into the expression for $s_{\nu}(t)$, eq 2, and initial conditions are known. Our goal, however, is different: we wish to derive phenomenological rate coefficients for use in global chemical modeling.

Chemical and Relaxational Subspaces. Physically it is clear that, after a short transitional period, the multiscale relaxational kinetics, which is governed by eq 1, is reduced to a much simpler situation, where only a few chemical relaxation processes exist. To see how this reduction takes place we notice that in general some of the eigenvalues of the kinetic relaxation operator $\hat{G}$ are much smaller than the remaining ones.¹⁸ In the most extreme situation, where no chemical transformations of the reactive complex are allowed or, equivalently, when the barriers separating different isomers and bimolecular species are infinitely high, there are N zero eigenvalues of the $\hat{G}$ operator, which correspond to N Boltzmann distributions for the different isomers. These distributions form an N -dimensional subspace of the total phase space, which we will call the chemical subspace. We will use the notation $|i\rangle$ to represent the i th isomer basis vector

$$|i\rangle = \frac{1}{\sqrt{Q_i}} \begin{cases} f_i^{(0)}(E) & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} \quad (16)$$

and Q_i is the i th isomer partition function

$$Q_i = \int dE f_i^{(0)}(E) \quad (17)$$

It is worth noting that for an arbitrary nonequilibrium $|f\rangle$ state the projection of $|f\rangle$ on the i th isomer basis vector $\langle i|f\rangle$ gives, up to the normalization factor $\sqrt{Q_i}$, the macroscopic concentration of the i th isomer

$$n_i = \sqrt{Q_i} \langle i|f\rangle \quad (18)$$

The subspace complementary to the chemical one will be called the relaxational subspace.

When the barriers separating different species are finite but high, there are still N eigenvalues that are much smaller than the rest because it still requires a long time for an isomer to overcome a barrier and to convert to another isomer or to bimolecular products. We call such eigenstates as chemically significant eigenstates¹³ (CSE) or shortly chemical eigenstates. All other eigenstates will be called internal energy relaxation eigenstates (IERE) or shortly relaxational ones.

The most important feature of the chemical eigenstates is that to a very good approximation they belong to the chemical subspace. However, they are not the same, and in the following discussion, it is important to understand the distinction between them. To illustrate the correlation between them we consider as an example the $C_3H_7 + O_2$ reaction. The potential energy surface for this reaction is shown in Figure 1. The details about the potential energy surface and about the densities and numbers of states for different wells and barriers can be found in the work of Goldsmith et al.²⁷ For our purposes, it is

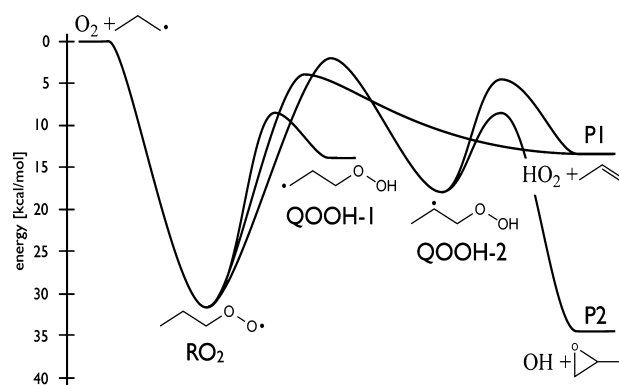


Figure 1. Potential energy surface for the $C_3H_7 + O_2$ reaction.²⁷

sufficient to note that there are three stable isomers for this reaction, which are denoted as follows: RO_2 as W_1 , $QOOH-2$ as W_2 , and $QOOH-1$ as W_3 . As a result, at low-to-moderate temperatures, there are three chemical eigenstates, whose eigenvalues are well separated from the eigenvalues associated with the relaxational eigenstates.

In Figure 2 the ratios of the chemical eigenvalues to the lowest relaxational one and the projections of the correspond-

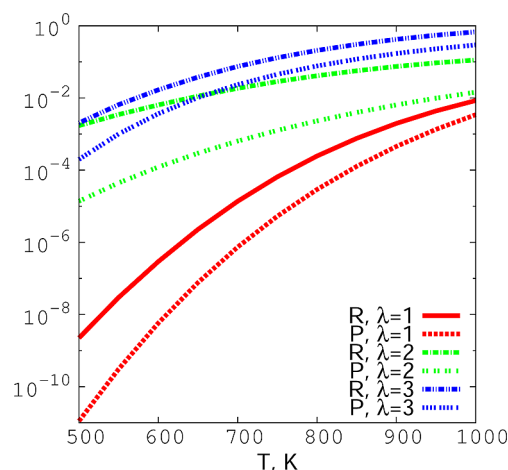


Figure 2. Eigenvalue ratio R and the eigenvector projection P for the $C_3H_7 + O_2$ reaction at 10 bar.

ing chemical eigenvectors onto the relaxational subspace are shown as functions of temperature. Here, the lowest relaxational eigenvalue will be called the energy relaxation limit. One can see that the projections of the chemical eigenvectors onto the relaxational subspace correlate with the corresponding eigenvalues and that these projections are very small when the chemical eigenvalues are well separated from the relaxational ones (this is seen to occur at low temperatures).

In Figure 3 selected eigenvectors are compared to the Boltzmann distributions in the corresponding wells for the $C_3H_7 + O_2$ reaction. Solid lines represent microscopic populations as functions of energy normalized with the same arbitrary factor in all wells (the lower index in the distribution notation corresponds to the well index). The distributions are shown only for those wells in which the populations are significant. For each chemical eigenstate (the upper row of panels) the dashed line represents the ratio of the eigenvector microscopic population to the Boltzmann distribution in the dominant well (the well with the maximal projection coefficient

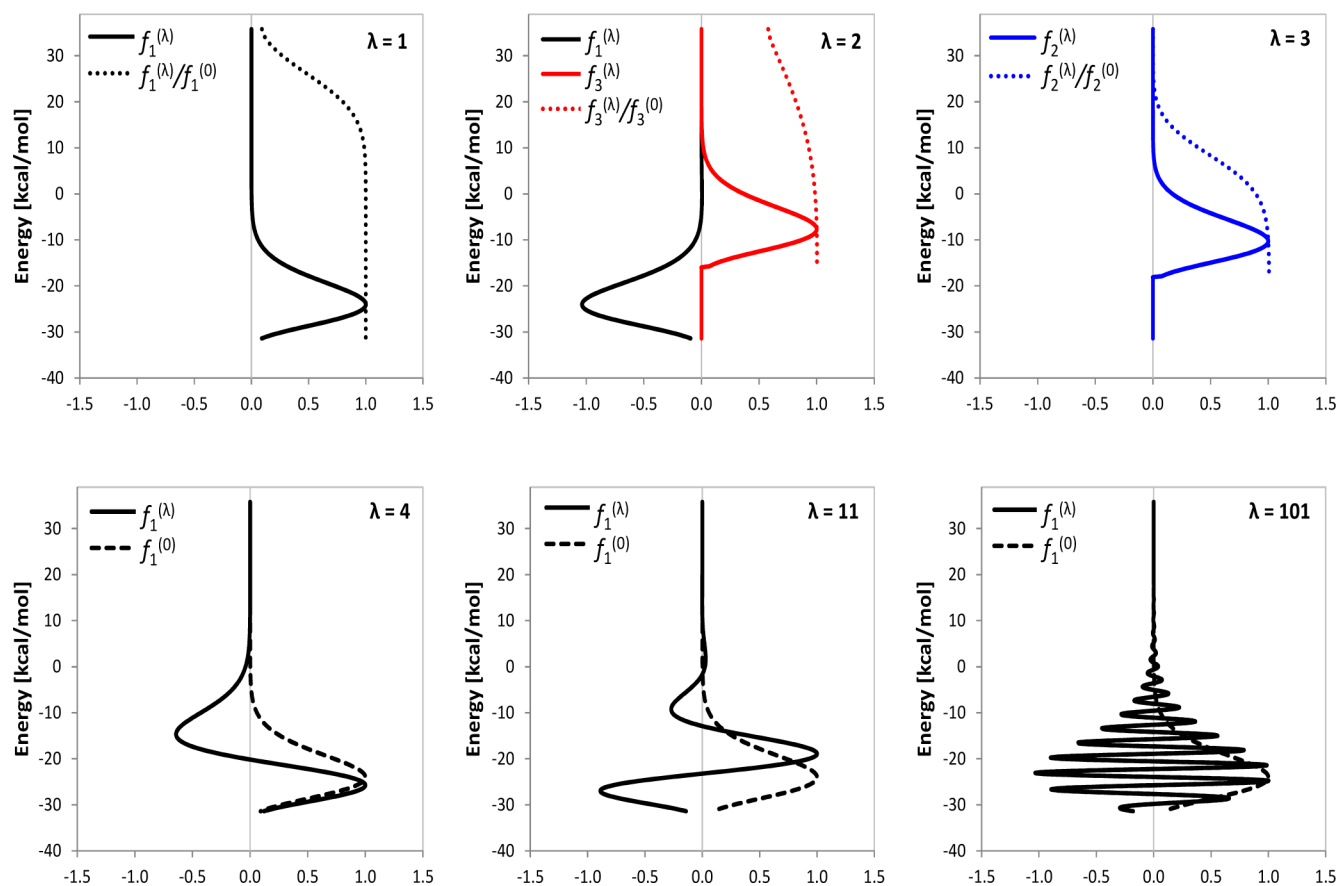


Figure 3. Selected eigenvectors for the $\text{C}_3\text{H}_7 + \text{O}_2$ reaction at the temperature of 600 K and pressure of 10 bar.

$\langle \lambda | i \rangle$). One can see that each chemical eigenstate closely mimics the corresponding Boltzmann distribution up until the dissociation threshold where it starts to deplete due to chemical reaction. In contrast, the relaxational eigenstates (the lower row of panels) are oscillatory and show little relation to the Boltzmann distributions shown with the dashed lines on those panels. The number of oscillations increases with increasing eigenvalue for these eigenstates.

Kinetic Phenomenology. We assume that the time scale for variation of the concentrations of bimolecular reactants, which enters into the factor $s_\nu(t)$, eq 2, is much longer than the slowest time scale associated with the relaxational eigenstates. Here, we will call the latter time scale the energy relaxation time scale. Then, for the relaxational eigenstates, after a short transitional period, the solution of eq 14 to a very good approximation is given by

$$f_\lambda = \frac{1}{\Lambda_\lambda} \sum_\nu s_\nu(t) p_\lambda^{(\nu)} \quad (19)$$

One can see that the expansion coefficients f_λ for the relaxational eigenstates do not depend on the initial conditions and respond instantaneously to the change in the external conditions.

The rate of formation of the μ th bimolecular products from an arbitrary reactive complex state $|f\rangle$ can be written as

$$r = \frac{1}{2\pi} \sum_i \int dE f_i(E) N_{i,\mu}^\#(E) / \rho_i(E) = \langle f | p^{(u)} \rangle = \sum_\lambda f_\lambda p_\lambda^{(\mu)} \quad (20)$$

The sum in this equation can be divided into two parts. The first part of the sum that refers to the chemical eigenstates will be considered later. It will be shown that it is related to the isomer-to-bimolecular reactions.

The second part of the sum in eq 20 that refers to the relaxational eigenstates has the form corresponding to bimolecular-to-bimolecular reactions and should be treated as such. Substituting eqs 2 and 19 into eq 20 one obtains the following expression for the phenomenological rate coefficient from the ν th bimolecular reactants to the μ th bimolecular products:

$$k_{\nu \rightarrow \mu} = \frac{1}{Q_\nu} \sum_\lambda \frac{1}{\Lambda_\lambda} p_\lambda^{(\mu)} p_\lambda^{(\nu)} \quad (21)$$

where the sum is taken over the relaxation eigenstates only and $p_\lambda^{(\nu)}$ is given by eq 15. We would like to emphasize that although the $k_{\nu \rightarrow \mu}$ rate coefficients are determined through the relaxation eigenvalues, the characteristic time scales they define are much longer. In fact one can view them as defining additional chemical time scales beyond those defined by the eigenvalues of $\hat{G}$.

The diagonal rate coefficient $k_{\nu \rightarrow \nu}$ describes the return back to bimolecular reactants, so that the overall reactants balance will be satisfied:

$$k_\nu^{(c)} = k_{\nu \rightarrow \nu} + \sum_{\mu \neq \nu} k_{\nu \rightarrow \mu} + \sum_i k_{\nu \rightarrow i} \quad (22)$$

where $k_\nu^{(c)}$ is the capture rate for the ν th bimolecular reactants

$$\begin{aligned}
 k_{\nu}^{(c)} &= \frac{1}{Q_{\nu}} \sum_i \sqrt{Q_i} \langle i | p^{(\nu)} \rangle \\
 &= \frac{1}{2\pi Q_{\nu}} \sum_i \int dE n_{i,\nu}^{\#}(E) e^{-E/T}
 \end{aligned}
 \quad (23)$$

and $k_{\nu \rightarrow i}$ are the bimolecular-to-isomer rate coefficients, eq 28, below.

To obtain the phenomenological rate coefficients for the isomer-to-isomer, bimolecular-to-isomer, and isomer-to-bimolecular reactions, one notes that the time-dependent concentrations n_i of different reactive complex isomers are in very good approximation associated with the expansion coefficients for the chemical eigenstates only and that the relaxation eigenstates have negligible contribution to n_i . The concentration of the i th isomer of the reactive complex in the arbitrary $|f\rangle$ state is given by eq 18. However, as was already pointed out, to very good approximation the chemical eigenstates belong to the chemical subspace, and correspondingly, the relaxational eigenstates are orthogonal to it, $\langle \lambda | i \rangle \simeq 0$ for relaxational λ . Thus, the expansion coefficients for the chemical eigenstates and the time-dependent concentrations n_i are coupled via the linear transformation:

$$n_i = \sqrt{Q_i} \sum_{\lambda} M_{i,\lambda} f_{\lambda}, \quad f_{\lambda} = \sum_i M_{\lambda,i}^{-1} n_i / \sqrt{Q_i} \quad (24)$$

where the square matrix M is given by

$$M_{i,\lambda} \equiv \langle i | \lambda \rangle = \frac{1}{\sqrt{Q_i}} \int dE f_i^{(\lambda)}(E) \quad (25)$$

Substituting eq 24 into eq 14 for the chemical eigenstates and comparing the result with the standard phenomenological chemical kinetics equations

$$\frac{dn_i}{dt} = -k_i n_i + \sum_{j \neq i} k_{j \rightarrow i} n_j + \sum_{\nu} k_{\nu \rightarrow i} n_A^{(\nu)} n_B^{(\nu)} \quad (26)$$

one obtains the expression for the phenomenological internal isomerization rate coefficients

$$k_{j \rightarrow i} = -\sqrt{Q_i/Q_j} (M \Lambda M^{-1})_{ij} \quad (27)$$

where by Λ we understand the diagonal matrix with the diagonal elements Λ_{λ} . This result is identical to that obtained using the long-time method from our previous work.¹⁵ It is also similar to the one of Bartis and Widom¹⁸ except that no orthogonality of the matrix M is assumed (the orthogonality of M is discussed further in the Discussion section below). Also, in the Bartis and Widom paper, the reactive complex was assumed to be conservative (no bimolecular reactants or products were included), whereas in the present case it is not.

The expression for the bimolecular-to-isomer rate coefficients is given by

$$k_{\nu \rightarrow i} = \frac{\sqrt{Q_i}}{Q_{\nu}} \sum_{\lambda} M_{i,\lambda} p_{\lambda}^{(\nu)} \quad (28)$$

where the $p_{\lambda}^{(\nu)}$ coefficients are given by eq 15, and the sum is taken over the chemical eigenstates only.

The diagonal matrix element $k_i = (M \Lambda M^{-1})_{ii}$ describes the overall loss for the i th isomer, for which chemical balance should be satisfied:

$$k_i = \sum_{j \neq i} k_{i \rightarrow j} + \sum_{\nu} k_{i \rightarrow \nu} \quad (29)$$

where $k_{i \rightarrow \nu}$ are the isomer-to-bimolecular rate coefficients, eq 30, below.

Because of the correspondence between the chemical eigenstates expansion coefficients and the time-dependent reactive complex isomer concentrations, eq 24, it is clear that the part of the sum in eq 20 that refers to the chemical eigenstates is related to the isomer-to-bimolecular reactions. Considering only that part of the sum in eq 20 and substituting eq 24 into it, one obtains the following expression for the isomer-to-bimolecular phenomenological rate coefficients:

$$k_{i \rightarrow \nu} = \frac{1}{\sqrt{Q_i}} \sum_{\lambda} M_{\lambda,i}^{-1} p_{\lambda}^{(\nu)} \quad (30)$$

where the $p_{\lambda}^{(\nu)}$ coefficients are given by eq 15 and the sum is once again taken over the chemical eigenstates only.

IV. DISCUSSION

Under the conditions of high temperatures and/or low pressures, some of the chemical eigenvalues may approach the energy relaxation limit. When this happens, the projection of the corresponding eigenvectors onto the relaxational subspace increases and eventually this projection becomes the dominant one. As a result, the rate calculation approach described in the previous section, which is implicitly based on the assumption that the projection of the chemical eigenvectors onto the relaxational subspace is negligible, must be modified.

Physically, however, the situation is clear. Generally speaking, the chemical eigenstates describe the process of equilibration between two groups of the isomers or between a group of the isomers and the bimolecular species.^{15–17} For an eigenvector describing equilibration with bimolecular products, the macroscopic populations for each isomer have the same sign. For an eigenvector that describes equilibration between two groups of isomers, the cumulative populations for those groups are generally approximately equal in magnitude and opposite in sign, at least at not too high a temperature. Thus, for the $C_3H_7 + O_2$ reaction, the first eigenvector describes equilibration between the $W_1 + W_3$ group and the bimolecular products, the second eigenvector describes equilibration between the W_1 and W_3 isomers, and the third eigenvector describes equilibration between the W_2 isomer and the bimolecular products, cf. Figure 3.

The situation where some of the chemical eigenvalues approach the energy relaxation limit implies that the equilibration between the corresponding groups of species occurs so rapidly that it cannot be separated from the energy relaxation processes. The corresponding eigenstates should then be considered as effectively relaxational ones and the effective number of species should be reduced. Those species that are in equilibrium with each other should be united and treated as one and the dimensionality of the effective chemical subspace should be reduced as well. Then, the expressions for the phenomenological rate coefficients given by eqs 27, 28, and 30 remain valid if one interprets the i th isomer to mean the united species g , whose basis vector $|g\rangle$ can be described as

$$|g\rangle = \frac{1}{\sqrt{Q_g}} \begin{cases} f_j^{(0)}(E) & \text{if } j \in g \\ 0 & \text{if } j \notin g \end{cases} \quad (31)$$

$$Q_g = \sum_{j \in g} Q_j \quad (32)$$

The matrix M , eq 25, should be modified accordingly.

The expression for the bimolecular-to-bimolecular rate coefficients, eq 21, remains formally unchanged. It is worth noting that when the dimensionality of the effective chemical subspace differs from the original one, part of the latter is implicitly included into the effective relaxational subspace. In this situation there may be nonzero contributions to the macroscopic populations of the reactive complex isomers from the relaxational eigenstates, which, as it has been shown [cf. eq 19], are associated with bimolecular species. These populational contributions can be interpreted as the concentrations of the isomers that are in equilibrium with some of the bimolecular species. Focusing on the specific ν th bimolecular species and using eqs 2, 13, 18, and 19, one obtains the following expression for the i th isomer equilibrium concentration:

$$\frac{n_i}{n_A^{(\nu)} n_B^{(\nu)}} = \kappa_{i,\nu} \frac{Q_i}{Q_\nu} \quad (33)$$

where the parameter $\kappa_{i,\nu}$,

$$\kappa_{i,\nu} = \frac{1}{\sqrt{Q_i}} \sum_{\lambda} \frac{1}{\Lambda_{\lambda}} \langle i|\lambda \rangle p_{\lambda}^{(\nu)} \quad (34)$$

should be close to unity if the i th isomer is in equilibrium with the ν th bimolecular species and to zero otherwise. The sum in eq 34 is taken over the relaxational eigenmodes only.

Any sound theory should guarantee that the expressions provided for the phenomenological rate coefficients satisfy the macroscopic detailed balance condition^{9,24}

$$\frac{k_{i \rightarrow j}}{k_{j \rightarrow i}} = \frac{Q_j}{Q_i} \quad (35)$$

For the bimolecular-to-bimolecular phenomenological rate coefficients the expression obtained here, eq 21, automatically satisfies the detailed balance condition. To show that the expressions for the internal isomerization and isomerization-to-bimolecular rate coefficients and inverse ones, eqs 27, 28, and 30, actually satisfy eq 35, one notes that different chemical eigenvectors are orthogonal to each other in the global phase space because they are the eigenvectors of the self-adjoint operator $\hat{G}$, cf. eq 11. As was pointed out earlier, to a very good approximation the chemical eigenvectors belong to the chemical subspace, which is a consequence of the separation of time scales for the chemical and energy relaxation processes. As a result the chemical eigenvectors are also, to a very good approximation, orthogonal to each other in the chemical subspace, and the matrix M , eq 25, which enters in eqs 27, 28, and 30, is orthogonal

$$M^{-1} \approx M^T \quad (36)$$

Thus, the macroscopic detailed balance condition, eq 35, is also satisfied for the phenomenological rate coefficients expressions of the internal isomerization and isomer-to-bimolecular and inverse reactions. It also shows that, in agreement with the general principles of nonequilibrium statistical mechanics,²⁸ deviations from the macroscopic detailed balance condition, eq 35, start to appear when there is no clear separation of time scales between the chemical relaxation processes and the collisional energy relaxation ones.

The relationship between the macroscopic detailed balance condition and the master equation is discussed in more detail elsewhere.^{9,24}

The kinetic operator $\hat{G}$ in eq 1 describes the time-evolution of the reactive complex without external supply from bimolecular reactions (it includes, however, bimolecular reactions as a sink). In some situations, it is important to know the result of such time-evolution for an arbitrary nonthermal distribution that can be formed as a result of some fast physical or chemical process. On the energy relaxation time scale such an evolution will lead to the formation of bimolecular products and different thermalized isomers, whose subsequent evolution will occur on a much longer time scale. The current formalism allows one to obtain closed expressions for the branching ratios between the different isomers and bimolecular products.

We assume that the initial nonequilibrium state vector $|f_0\rangle$ is described in terms of the expansion coefficients $f_{0,\lambda}$, eq 13. On the energy relaxation time scale the expansion coefficients for the chemical eigenstates do not change and the population fraction $n^{(i)}$ that contributes to the formation of the i th isomer is given by [cf. eq 18]

$$n^{(i)} = \sum_{\lambda} f_{0,\lambda} n_{\lambda}^{(i)}, \quad n_{\lambda}^{(i)} \equiv \sqrt{Q_i} \langle \lambda | i \rangle \quad (37)$$

where the sum is taken over the chemical eigenstates only.

To obtain the fraction of the initial population $n^{(\nu)}$ that goes to the ν th bimolecular products, one notes that the rate of formation of the ν th bimolecular products is given by eq 20 and is related to the relaxational eigenstates. The time-evolution of the expansion coefficients f_{λ} is described by eq 14 with zero source term, $f_{\lambda}(t) = f_{0,\lambda} \exp(-\Lambda_{\lambda} t)$. By substituting eq 13 into eq 20 (only relaxational eigenstates are considered) and integrating it over time, one obtains the following expression for $n^{(\nu)}$:

$$n^{(\nu)} = \sum_{\lambda} \frac{1}{\Lambda_{\lambda}} f_{0,\lambda} p_{\lambda}^{(\nu)} \quad (38)$$

where the $p_{\lambda}^{(\nu)}$ coefficients are given by eq 15 and the sum is taken over the relaxational eigenstates only.

Equation 38 can be rewritten in a more transparent form using the following property of an arbitrary eigenstate $|\lambda\rangle$,

$$\Lambda_{\lambda} n_{\lambda} = \sum_{\nu} p_{\lambda}^{(\nu)} \quad (39)$$

where n_{λ} denotes the overall reactive complex population associated with the eigenstate $|\lambda\rangle$ [cf. eq 18]

$$n_{\lambda} = \sum_i \sqrt{Q_i} \langle i | \lambda \rangle \quad (40)$$

Equation 39 is a consequence of the overall reactive complex population balance. By substituting eq 39 into eq 38, one obtains

$$n^{(\nu)} = \sum_{\lambda} f_{0,\lambda} n_{\lambda} \frac{p_{\lambda}^{(\nu)}}{\sum_{\mu} p_{\lambda}^{(\mu)}} \quad (41)$$

For qualitative analysis, it is convenient to take as an initial nonequilibrium distribution a δ -function in energy in a particular well. Then the fractions of the bound species and bimolecular products calculated according to eqs 37 and 38 represent the probabilities for the reactive complex with the

specific initial energy and in the specific initial well to be either deactivated in one of the wells or to decompose into the corresponding bimolecular products. As an illustration, in Figure 4, the corresponding probabilities are shown as

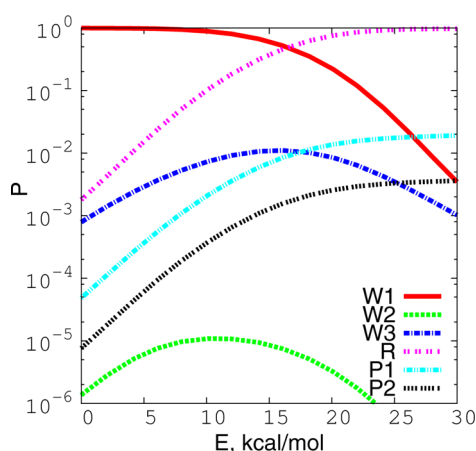


Figure 4. Probabilities for the reactive complex to be captured in different wells or dissociate to different bimolecular products for the $\text{C}_3\text{H}_7 + \text{O}_2$ reaction at 600 K and 1 bar.

functions of energy for the reaction $\text{C}_3\text{H}_7 + \text{O}_2$ at 600 K and 1 bar. The initial distribution corresponds to the RO_2 isomer (W_1 well) at some energy E relative to the ground state of $\text{C}_3\text{H}_7 + \text{O}_2$. One can see that at energies below 10 kcal/mol the reactive complex most probably deactivates in the same W_1 well, while at energies above 20 kcal/mol it dissociates to $\text{C}_3\text{H}_7 + \text{O}_2$ (R on Figure 4). Probabilities to be captured in the other wells or to dissociate to the different products are small at all energies.

In the earlier approach to the multiwell problem, which was developed by some of us,^{11,13,15} the pair of bimolecular reactants was explicitly included into the master equation. Assuming that one of the bimolecular reactants is in great excess over the other, and providing the rate equation for the second bimolecular reactant allowed one to include the concentration of the second bimolecular reactant into the dynamical phase space. This pair of bimolecular reactants can then be viewed as an additional isomer, whose equilibration with bimolecular products provides one additional chemical eigenstate. The depth of the well for this additional isomer is related to the concentration of the excess bimolecular reactant: smaller concentration corresponds to a deeper well. Such an approach gives phenomenological rate coefficients that depend slightly on the concentration of the excess reactant. This dependence, however, is negligible in almost all imaginable situations. The current approach then formally corresponds to the limit of zero-concentration for the excess bimolecular reactant.

It is somewhat surprising, at least for those familiar with our previous work, that the relaxational modes contribute to some of the rate coefficients. In fact, they completely determine them for reactions that have both bimolecular reactants and bimolecular products. There is some analogy between our current approach and the chemical-activation methodology used to determine such rate coefficients.^{29,30} Consider a single-well system with two sets of bimolecular fragments, both of which are taken to be sinks. The problem has only one chemical eigenstate, corresponding to thermal dissociation from

the well. We will call one of the bimolecular configurations the reactants, R, and the other the products, P, and assume that the initial distribution in the well is given by $(2\pi Q_R)^{-1} N_R^\#(E) e^{-E/T}$ where Q_R is the partition function of the reactants, eq 3, and $N_R^\#(E)$ is the number of states for the transition state separating the reactants R from the well, which is the reactive complex energy distribution immediately after the capture [cf. eq 23]. We then integrate the master equation in time until the accumulated macroscopic populations of the reactants, products, and the well are no longer changing perceptibly in time, which happens to be on the energy relaxation time scale. Such a situation will occur as long as the chemical eigenvalue is much smaller in magnitude than the ones describing energy relaxation. Note that the time evolution we consider is governed completely by the relaxational eigenstates. The rate coefficients for the $\text{R} \rightarrow \text{well}$ and $\text{R} \rightarrow \text{P}$ reactions are then equal to fractional macroscopic populations accumulated in the well and in the products, respectively.

Thus, the $\text{R} \rightarrow \text{P}$ reaction rate coefficient is related to the part of the activated reactive complex population that is associated with the relaxational eigenstates, eqs 38 and 41. The $\text{R} \rightarrow \text{well}$ reaction rate coefficient is related to the part of the activated reactive complex population that is associated with the chemical eigenstate and therefore evolves slowly with time, eq 37. In fact, the final distribution formed in the well is proportional to the chemical eigenvector, which subsequently thermally dissociates on the chemical time scale.^{9,15}

V. CONCLUDING REMARKS

Equations 21, 27, 28, and 30 provide simple expressions that can be used for calculating the full set of phenomenological rate coefficients for a given multiple-well, multiple-bimolecular-species chemical system. These expressions have been incorporated into a new master equation code that is under development in our group. Sample calculations indicate that the results from this code are fully consistent with those derived from our earlier implementation based on the formulation that directly treats the dynamics of the bimolecular reactants. These new expressions (and the underlying derivations) provide a useful alternative perspective on the relationship between the microscopic population dynamics described by the master equation and the phenomenological rate constants employed in global chemical modeling.

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Notes

The authors declare no competing financial interest.

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